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Forecasting crude oil prices: A Gated Recurrent Unit-based nonlinear Granger Causality model

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ABSTRACT

Crude oil prices substantially impact the global economy, national security, and international politics. Accurate forecasting of crude oil prices and understanding the causality mechanism are crucial for policymakers, investors, and academia. Given the multitude of influence factors and oil prices' complex nonlinearity, we propose a novel Gated Recurrent Unit-based non-linear Granger Causality (GRU-GC) model to provide precise oil price forecasting and causality determination. First, we collect numerous potential influence factors, including supply and consumption, global economic development, financial factors, and the energy market. We then harness the powerful predictive capabilities of the GRU, coupled with the insightful interpretation offered by the linear Granger model. This enables point prediction, probability density estimation, and causal inference simultaneously. The experimental results on the Brent and West Texas Intermediate (WTI) oil data sets show a significant superiority in multi-step-ahead forecasting. Furthermore, the unexpected variability of oil prices spurred by the COVID-19 pandemic validates the need and efficacy of integrating a probability density function for robust prediction. Additionally, our model reveals the non-linear Granger causality that differentiates Brent and WTI oil prices.

1. Introduction

As a pivotal commodity, crude oil profoundly influences national security, global economy, energy conservation, international politics, and historical events (Henriques & Sadorsky, 2011; Mei, Ma, Liao, & Wang, 2020; Omar & Lambe, 2022; Shao & Hua, 2022). According to BP's Statistical Energy Outlook, oil will continue to play a significant role in the global energy system, with the world consuming between 70–80 Mb/d in 2035.[2] Forecasting crude oil prices and understanding their patterns is essential for policy adjustments (Abdollahi & Ebrahimi, 2020) and monitoring supply–demand changes (Känzig, 2021).

Crude oil price variability is largely driven by the fundamental forces of supply and demand (Zhao, Li, & Yu, 2017). However, beyond these core factors, various other elements influence oil prices at different frequencies. Global economic indicators, such as economic growth and technological development, also play a role in shaping oil prices (Pan, Wang, Wu, & Yin, 2017). Financial factors, including market speculation (Working, 1960), as well as the stability and interconnectedness of the global banking system (Zhang, Gabauer, Gupta and Ji, 2024), significantly impact oil price dynamics. In the energy market, the production of substitute commodities such as natural gas and clean energy (Sadorsky, 2022) exerts a substitution effect, indirectly leading to oil price fluctuations. Moreover, external shocks and uncertainty factors, such as geopolitical risks (Jin, Zhao, Bu, & Zhang, 2023; Li, Liang, Chen, & Umar, 2022), economic policy uncertainty (Kang & Ratti, 2013), further contribute to the variability of oil prices in unique ways. Given this complexity, accurately capturing the relationships between crude oil prices and these influencing factors is crucial for providing insightful guidance to policymakers. Identifying the most relevant variables as predictors not only helps to clarify these intricate relationships but also reduces the risk of overfitting and ultimately improves forecast accuracy (Luo, Klein, Walther and Ji, 2024).

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[2] https://www.bp.com/en/global/corporate/energy-economics/energy-outlook.html.

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Crude oil price forecasting has attracted significant attention from scholars and practitioners, resulting in two primary prediction methods. The first stream comprises traditional econometric and statistical methods that analyze the impact of various predictors on crude oil prices (Ji, Bouri, & Roubaud, 2018; Zhang, Ji, & Kutan, 2019). These include the linear Granger model (Yu, Li, Tang and Wang, 2015), autoregressive integrated moving average (ARIMA) (Xiang & Zhuang, 2013), generalized autoregressive conditional heteroskedastic model (GARCH) (Ghani, Zhu, Ma, & Ghani, 2024; Lin, Xiao, & Li, 2020), vector autoregression model (VAR) (Wang, Wu, Zhang, & Ji, 2023), heterogeneous autoregressive (HAR) model and its variants (Luo, Klein, Walther, & Ji, 2021), and structural models (Wang & Sun, 2017). These models can capture time-varying variability and thus improve forecasting accuracy. However, as they assume the data to be stationary, regular, and linear, they cannot accurately model time series that are complex, irregular, and nonlinear (Yu, Wang and Tang, 2015).

The second crude oil price forecasting approach uses machine learning models, which address traditional model limitations by detecting complex patterns. These models include support vector machines (SVM) (Yu, Xu, & Tang, 2017), artificial neural networks (ANN) (Baruník & Malinská, 2016; Li, Jiang, Li, & Wang, 2021), recurrent neural networks (RNN) (Fischer & Krauss, 2018), and long short-term memory neural networks (LSTM) (Zhang & Hong, 2022). To solve the long-term dependence of RNN and the disappearance of gradients, the gated recurrent unit (GRU) has been introduced as a more sophisticated machine learning model (Cho, van Merrienboer et al., 2014). Compared with LSTM, GRU has a more compact architecture and better computational efficiency (Bin & Jun, 2021). Regarding energy price prediction and analysis, the GRU models have exhibited robust potential and achieved satisfying results. Li and Wang (2020) proposed a stochastic GRU model, incorporating a stochastic time intensity function, to predict the daily closing prices of West Texas Intermediate (WTI) and Brent crude oil. Niu, Xu and Liu (2021) developed a hybrid attention-based GRU model to enhance the forecasting accuracy of Brent oil prices. Bin and Jun (2021) employed a random deep bidirectional GRU model to refine the prediction accuracy of international crude oil prices. Zhang, Luo, Wang, and Liu (2023) introduced a hybrid GRU neural network based on decomposition-reconstruction methods to forecast WTI daily closing oil prices. Compared with econometric models, machine learning methods can model complex characteristics such as nonlinearity and variability. However, these advanced machine learning methods can encounter issues such as overfitting or sensitivity to parameter selections (Guo, Zhang, Liao, Chen, & Zeng, 2021; Wang et al., 2018).

The existing crude oil variability forecasting literature has identified two main limitations. Firstly, many forecasting methods used in previous studies incorporate all influence factors as dependent variables in the prediction model without explicitly identifying the core variables relevant to the research context. This ineffective variable selection can lead to suboptimal prediction outcomes due to the high volume of intensive data and numerous indicators involved. Secondly, a significant portion of prior research focuses solely on generating point predictions for crude oil prices, overlooking the importance of analyzing the entire price distribution. Examining the distribution of oil prices can provide valuable insights into uncertainty, particularly for nonlinear data.

To overcome the above two limitations, this study proposes a novel Gated Recurrent Unit-based non-linear Granger Causality model (GRU-GC) specifically designed for time series data analysis. By leveraging the strengths of Granger causality and the GRU model, the GRU-GC model aims to achieve both accurate predictions and interpretability. The integration of Granger causality allows for identifying relationships among variables, while the GRU model captures non-linear interactions among multiple influence factors to enhance prediction accuracy and robustness. To address variable selection in the presence of non-linear relationships, the GRU-GC model incorporates L_1 regularization for inter-group penalties and L_2 regularization for intra-group penalties, effectively filtering out irrelevant variables that have no Granger-causal impact on crude oil prices. Furthermore, by incorporating predicted distributions, it becomes possible to derive interval estimations for oil prices, offering policymakers more nuanced guidance.

We perform a rigorous empirical analysis of Brent and WTI data sets. The GRU-GC model demonstrates superiority over seven competing forecasting models for accuracy. Furthermore, we identify two key findings. Firstly, due to its unique probabilistic forecasting capabilities, the GRU-GC model effectively captures the significant variability in oil prices triggered by the unexpected COVID-19 pandemic event in February 2020. Secondly, the model distinguishes causality between Brent and WTI oil price fluctuations. We observed that European-related factors predominantly influence Brent oil prices, while U.S.-related factors take precedence for WTI data, aligning with existing literature. This study's significant contributions are as follows:

- We contribute to the existing literature by proposing a variable selection method that can effectively accommodate a wide range of factors influencing crude oil prices. Our approach takes into account various aspects, including supply and demand dynamics, global economic development, financial factors, energy market indicators, and external shocks to provide a comprehensive and holistic view of the crude oil market. In contrast to many existing models that consider only a limited set of factors, our proposed GRU-GC model incorporates a group regularization method. This enables efficient variable selection and allows us to capture higher-order interaction effects among variables that are often overlooked by linear models. As a result, our model offers accurate forecasts while mitigating the risk of overfitting.
- Our model provides valuable insights into the relative significance of influence factors contributing to price variability through Granger causality analysis. Unlike traditional deep learning models, our approach incorporates Granger causality explanations, enabling the identification and prioritization of core variables that play a crucial role in driving crude oil prices. This goes beyond the scope of mere prediction, as our model helps uncover the underlying causal mechanisms behind price fluctuations. This deeper understanding of the magnitude of price fluctuations is invaluable for policymakers who need to make informed strategic decisions in the dynamic crude oil market.
- Our study goes beyond traditional models by establishing the probability density prediction of crude oil prices. Rather than providing point estimations, our approach forecasts the distribution of crude oil prices at each point in time. This enables us to capture and account for the inherent uncertainty that is characteristic of crude oil prices. By considering the full probability density, our predictions become more robust and reliable, providing policymakers with a comprehensive understanding of the potential range of crude oil price outcomes.

The structure of this paper can be summarized as follows. Section 2 outlines the related work on influence factors and the hybrid forecasting approaches. Section 3 provides the forecasting framework consisting of a novel GRU-GC model adapting to point or distribution prediction. Section 4 describes the empirical analysis, consisting of the data and influence factors for Brent crude oil prices. Section 5 illustrates the performance of the proposed model and comparative analysis with its competing models. Finally, our conclusion is presented in Section 6.

2. Related work

In this section, we review the literature on the factors influencing crude oil prices and examine hybrid forecasting approaches used for crude oil price prediction.

2.1. Influence factors

In examining the factors influencing crude oil prices, researchers have extensively studied various aspects, including supply and demand dynamics, global economic growth, financial factors, energy markets, external shocks, and uncertainty factors. Specifically, supply and demand dynamics play a fundamental role in the variability of oil prices. Previous research has demonstrated the significant influence of supply and consumption on oil price fluctuations (Bernabe, Martina, Alvarez-Ramirez, & Ibarra-Valdez, 2004). For example, Kaufmann (2011) attributed oil price changes in 2007–2008 to changes in Chinese demand and non-OPEC producer supply. Wang and Sun (2017) employed structural equation modeling to identify determinants of oil prices and found that oil demand is relatively inelastic to price fluctuations.

Global economic conditions, which reflect the economies, profoundly impact oil prices, particularly on the demand side. Studies indicate that the economic growth of major oil-consuming countries significantly affects oil demand (Lu, Li, Chai, & Wang, 2020). Additionally, global economic events such as recessions, financial crises, and geopolitical tensions can impact oil demand and price fluctuations.

Financial factors, such as market sentiment, speculation, and psychological expectations, also play a crucial role in determining oil prices (Wei, Liu, Lai, & Hu, 2017). These factors are measured using data from stock markets and monetary markets. Studies have shown that incorporating stock market variability enhances the predictive accuracy of oil variability (Degiannakis & Filis, 2017), and there are time-evolving patterns of variability spillover across multiple periods (Hou, Li, & Wen, 2019). The relationship between oil price shocks and exchange rates has also been explored, revealing heterogeneous effects at different time horizons (Kocoglu, Kyophilavong, Awan, & Lim, 2022). Furthermore, a significant correlation between gold and oil prices has been observed in Zhang and Wei (2010).

The energy market includes commodities such as natural gas, coal, and steam coal, which are traded on spot and futures markets (Geng, Ji, & Fan, 2017; Jin et al., 2023). The prices of these energy commodities are interconnected, with variations in one commodity influencing the demand and price of others. For example, if the price of crude oil increases, consumers may switch to natural gas or coal, increasing demand for those commodities and their prices. Additionally, interdependence exists between these commodities, as increased crude oil prices could increase demand for natural gas or coal as substitutes, affecting their prices.

The oil market is also sensitive to external shocks and uncertainty factors. Baumeister and Hamilton (2019) identify four primary types of shocks that drive these changes: oil supply shocks, economic activity shocks, oil consumption demand shocks, and oil inventory demand shocks. Each type of shock uniquely affects the market. For instance, oil supply shocks, such as geopolitical events or natural disasters, disrupt oil availability and impact prices by altering the supply–demand balance. Oil consumption demand shocks result from changes in oil consumption patterns due to technological advancements or shifts in consumer behavior. Scholars also emphasize the importance of economic uncertainty in predicting oil prices and volatility. For instance, Kang and Ratti (2013) note that policy uncertainty explains oil volatility beyond standard macroeconomic factors. Wei et al. (2017) argue that EPU indices offer more predictive power than demand-supply or speculation indices for daily oil volatility and Li et al. (2022) expand the analysis to 15 uncertainty indices.

Given the large volume of data and numerous indicators involved, although the effective variable selection methods are used to achieve robust prediction outcomes, for instance, Luo, Ma, Wang and Wu (2024) employed dimension reduction techniques on a large number of predictors to enhance the accuracy of oil price variability forecasting, they cannot investigate the relative importance of these influencing factors. The proposed model in this study can help decision-makers further understand which factors impact oil price variability more than others.

2.2. The hybrid forecasting approach

The hybrid models combine the strengths of econometric/statistical models and machine learning models to forecast crude oil prices, providing both precision and interpretability. Studies have experimentally proven that such hybrid approaches effectively improve forecasting performance. For example, Chai, Xing, Zhou, Zhang, and Li (2018) used the time-varying parameter structure in a time series model to predict crude oil prices, achieving notable accuracy. Abdollahi (2020) proposed a hybrid model integrating complete ensemble empirical mode decomposition, SVM, particle swarm optimization, and Markov-switching GARCH to capture nonlinearity and variability in the time series. Experimental findings demonstrated that the hybrid model outperformed other benchmark models. Similarly, Huang and Deng (2021) combined LSTM with a moving-window strategy and variational mode decomposition to enhance the precision of crude oil price forecasting. Empirical analysis consistently indicates that hybrid forecasting methods outperform individual methods in terms of accuracy. This enhanced performance is primarily attributed to the hybrid approach's ability to leverage the strengths of different models while compensating for their weaknesses.

Despite the promising results of hybrid models, most existing research on crude oil forecasting focuses exclusively on generating point predictions. However, analyzing the distribution of oil prices can yield valuable insights regarding uncertainty, particularly for nonlinear oil price data. By utilizing predicted distributions, it is possible to derive interval estimations for oil prices, thereby providing more comprehensive guidance to policymakers. In summary, Table 1 concludes the characteristics of some recent representative research studies, along with a comparison to our developed method in terms of modeling nonlinearity, selecting variables, and estimating probability distribution.

3. Proposed forecasting framework

In this section, we present a GRU-GC framework, as illustrated in Fig. 1. We begin by providing a brief overview of linear and nonlinear Granger causality and their relevance to our model. We then describe how the GRU-GC model captures the nonlinearity among temporal variables. The GRU-GC model offers both point and distribution predictions for crude oil prices, allowing for a comprehensive understanding of future outcomes. Furthermore, we explain the regularization techniques employed in the model, enabling us to assess the importance of each variable in forecasting the target variable.

We consider a multivariate time series denoted as $X_t = [X_t^1, X_t^2, \dots, X_t^D]^T \in \mathbb{R}^D$, where D represents the number of variables that can be observed at each time step t. The time series variables include supply and demand, global economic development, financial factors, energy market, and external shocks. These variables are known to exhibit nonlinear and complex behaviors, which pose challenges in accurately describing their patterns and forecasting future trends (Abdollahi, 2020; Li, Zhu, & Wu, 2019). Therefore, this study aims to develop a model that captures the underlying mechanisms of crude oil pricing and provides precise forecasts for spot prices.

3.1. Linear Granger causality

Granger causality analysis is a statistical method that identifies causal relationships in multivariate time series data (Granger, 1969). The original formulation of Granger causality is rooted in the framework of conditional probability distributions. It is defined as a measure of whether the past values of one time series, X_t^i, help predict future values of another series, X_t^j, beyond what is possible using just the past values of X_t^j. Formally, X_t^i does not Granger-cause X_t^j if for all $k > 0$, the following conditional probability equality holds (Osińska, 2011):

$$F(X_{t+k}^j | \Omega_t) = F(X_{t+k}^j | X_t^i, \Omega_t \setminus X_t^i) \tag{1}$$

Table 1
Comparisons with some representative research.

Study	Nonlinearity	Variable selection	Probabilistic estimation
Lu et al. (2020)	✓	✓	×
Li and Wang (2020)	✓	×	✓
Niu, Xu et al. (2021)	✓	×	×
Hao, Feng, Yuan, Sun, and Li (2022)	✓	✓	×
Wang, Li, and Li (2022)	✓	×	✓
Ours (GRU-GC)	✓	✓	✓

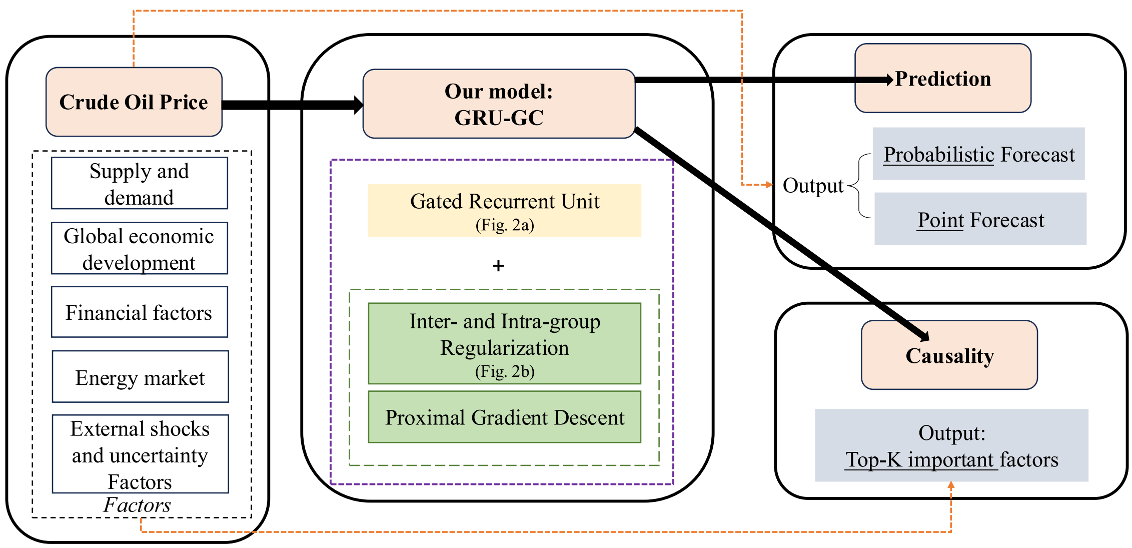


Fig. 1. Framework of the proposed GRU-GC.

where F denotes the conditional probability distribution, and Ω_t represents the full information set available at time t. If this equality does not hold, we say that X_t^i Granger-causes X_t^j, as it contributes unique information to forecasting X_t^j.

This definition is model-free and reflects the notion that causality implies the ability of one variable to provide additional predictive power for another. Commonly, the Vector Autoregression approach is one technique that assumes linear relationships between variables, allowing for testing Granger causality in the context of linear models (Lütkepohl, 2005). Specifically, the linear Granger causality test assumes

$$x_t = \sum_{k=1}^{K} \mathbf{A}_k x_{t-k} + \varepsilon_t \tag{2}$$

where x_t represents a specific realization of X_t. The maximum lag allowed in the model is denoted by K, indicating how many past observations are considered for prediction, and ε_t represents the error term, assumed to have a zero mean. The coefficient matrix $\mathbf{A}_k \in \mathbb{R}^{D\times D}$ represents the linear impact of lag k on the future evolution of the multivariate time series.

Linear Granger causality examines the causal relationship between two time series variables, X_t^i and X_t^j. If the coefficients $\mathbf{A}_{1:K}^{i,j} = (A_1^{i,j}, \ldots, A_K^{i,j})$ are jointly significant, it suggests that X_t^i Granger causes X_t^j, and vice versa. The omission of past values of X_t^i would decrease the forecast accuracy of X_t^j if indeed X_t^i Granger causes X_t^j. In cases where the time series dimension is high, the group lasso regression technique can be employed to solve Eq. (2) (Lozano, Abe, Liu, & Rosset, 2009).

$$\min_{\mathbf{A}_1,\ldots,\mathbf{A}_K} \sum_{t=1}^{T} \left\| x_t - \sum_{k=1}^{K} \mathbf{A}_k x_{t-k} \right\|_2^2 + \lambda \sum_{i,j} \left\| \left(A_1^{i,j}, \ldots, A_K^{i,j}\right) \right\|_2 \tag{3}$$

In Eq. (3), the L_2 norm, denoted by $\|\cdot\|_2$, is used in conjunction with a hyperparameter λ to control the extent of the penalty imposed on the coefficients. While the linear Granger causality test is commonly used, it may not adequately capture the intricate patterns of time series evolution due to its limited ability to capture interactive relationships between variables, as discussed by Yu, Li et al. (2015).

3.2. Gated recurrent unit-based Granger causality model

Previous research on oil price forecasting has predominantly focused on linear Granger causality, overlooking the potential nonlinear causal interactions. To address this limitation, several studies have incorporated nonlinear models to approximate a mapping function $f(\cdot)$ that captures the nonlinear causal relationships among variables:

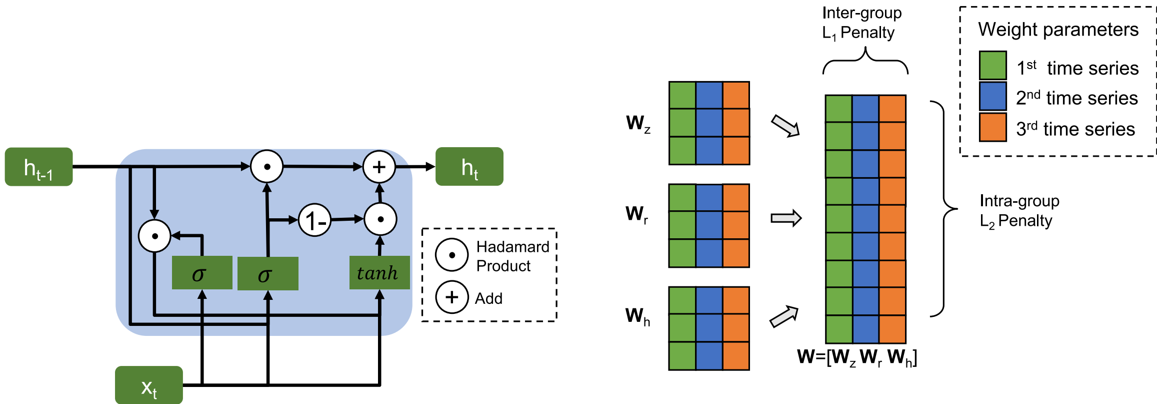
$$x_t = f(x_{<t}^1, \ldots, x_{<t}^D) + \varepsilon_t \tag{4}$$

where $x_{<t}^d$ represents the preceding t values of the dth series, and the noise term ε_t follows a Gaussian distribution with a zero mean. The function $f(\cdot)$ is commonly implemented using neural networks, such as RNN (Cho, Van Merriënboer, Bahdanau and Bengio, 2014) or LSTM (Hochreiter & Schmidhuber, 1997). However, these models often lack interpretability and do not support statistical tests of time series coefficients, which is essential for understanding Granger causality effects. To overcome this limitation, we propose a novel approach called GRU-GC, building on the framework proposed in Tank, Covert, Foti, Shojaie, and Fox (2021).

For two times series, i and j, if time series i does not nonlinearly Granger cause time series j, the following equation applies:

$$f_j(x_{<t}^1, \ldots, x_{<t}^{i-1}, x_{<t}^i, x_{<t}^{i+1}, \ldots, x_{<t}^D) = f_j(x_{<t}^1, \ldots, x_{<t}^{i-1}, x_{<t}^{i+1}, \ldots, x_{<t}^D) \tag{5}$$

where the component functions $f_j(\cdot)$ within the function $f(\cdot)$ are used to predict the values of variable x_t^j. These functions are implemented using a GRU, which is more resilient to gradient vanishing and exploding issues in sequential data modeling than RNN. Additionally, GRUs offer performance comparable to LSTM but are more computationally



(a) The structure of GRU.

(b) The visualization of L_1 and L_2 regularizations on parameters of GRU-GC.

Fig. 2. The illustration of GRU and regularization techniques used in GRU-GC.

efficient due to fewer parameters. Fig. 2(a) illustrates a GRU structure, and the mechanism of the GRU is described as follows:

$$\begin{aligned} z_t &= \text{sigmoid}\left(\mathbf{W}_z x_t + \mathbf{U}_z h_{t-1} + b_z\right) \\ r_t &= \text{sigmoid}\left(\mathbf{W}_r x_t + \mathbf{U}_r h_{t-1} + b_r\right) \\ \tilde{h}_t &= \tanh\left(\mathbf{W}_h x_t + \mathbf{U}_h\left(r_t \odot h_{t-1}\right) + b_h\right) \\ h_t &= z_t \odot h_{t-1} + \left(1 - z_t\right) \odot \tilde{h}_t \end{aligned} \tag{6}$$

where z_t, r_t, $\tilde{h}_t$ represent update gates, reset gates, and the candidate update gates, respectively, with h_t denoting the hidden state. The coefficients matrices $\mathbf{W}_z, \mathbf{W}_r, \mathbf{W}_h$ are input-to-hidden parameters, while $\mathbf{U}_z, \mathbf{U}_r, \mathbf{U}_h$ are hidden-to-hidden parameters. Bias vectors are given by b_z, b_r, and b_h. The sigmoid function, denoted as sigmoid$(\cdot)$, and the hyperbolic tangent function, denoted as $\tanh(\cdot)$, are two activation functions used to introduce non-linearity and facilitate the learning of complex data patterns.

The GRU model's interpretability is limited due to its complex architecture, including multiple hidden layers. This complexity makes it challenging to understand the specific mechanisms contributing to accurate predictions, particularly in identifying the input variables that significantly impact the output variable predictions. The lack of interpretability can hinder gaining insights into the underlying causal relationships and their relative importance.

For a GRU model, $\mathbf{W} = [\mathbf{W}_z, \mathbf{W}_r, \mathbf{W}_h]^T$ represents the input-to-hidden parameters. The proposed GRU-GC model enhances the traditional GRU by integrating a Granger causality test and applying both L_1 and L_2 regularizations to these parameters, which helps identify relevant variables and mitigate the influence of non-causal ones. Fig. 2(b) illustrates these penalties. The regularization term, denoted $\mathcal{L}_p$, is specified as follows.

$$\mathcal{L}_p = \lambda \sum_{i=1}^{D} \|\mathbf{W}^{:,i}\|_2 \tag{7}$$

where the L_1 norm, indicated by the summation of positive terms, represents the inter-group penalty across all variables. It encourages sparsity by reducing the parameters of variables that have little or no Granger causality effect on the target variable. The L_2 norm represents the intra-group penalty for a single variable. It penalizes the individual parameters within a variable, encouraging them to be small. Note that the sparsity-inducing regularizers, such as LASSO and group LASSO, have been used in linear Granger causality estimation under high-dimensional settings (Basu & Michailidis, 2015; Lozano et al., 2009). Although these techniques are also used in selecting neural networks' architecture (Alvarez & Salzmann, 2016; Louizos, Ullrich, & Welling, 2017), they are not proposed for interpreting the relationship between the input time series and the target. The hyper-parameter λ controls the degree of penalty imposed on the coefficients. By adjusting the value of λ, the model can regulate the importance of each variable in the prediction process.

3.3. Point and probabilistic forecasting

The GRU-GC model offers two approaches for forecasting crude oil prices: point forecasting and probabilistic forecasting. In the point forecasting approach, the objective is to estimate a single value for the target variable. This is achieved by minimizing the loss function $\mathcal{L}_1$, which is defined as the sum of squared differences between the actual and predicted values of the target variable, along with the penalty term $\mathcal{L}_p$. The objective function of the point forecasting approach is expressed as:

$$\begin{aligned} \mathcal{L}_1 &= \min \sum_{t=1}^{T-1} (x^j_{t+1} - \hat{x}^j_{t+1})^2 + \mathcal{L}_p \\ &= \min \sum_{t=1}^{T-1} (x^j_{t+1} - \hat{x}^j_{t+1})^2 + \lambda \sum_{i=1}^{D} \|\mathbf{W}^{:,i}\|_2 \end{aligned} \tag{8}$$

where x^j_{t+1} represents the actual value of the target series at time step $t+1$, while $\hat{x}^j_{t+1}$ represents the predicted value. The hidden state at time step t is denoted as h_t, and the predicted target series X^j_t at time step $t+1$ is determined by a feed-forward neural network $g_j(\cdot)$, as illustrated in Fig. 3.

In addition to point forecasting, the GRU-GC model offers a probabilistic approach to forecasting crude oil prices. Instead of estimating a single value, this approach models the target series as a Gaussian distribution at each time step. The model outputs the mean and standard deviation of the distribution, which provide information about the central tendency and spread of the predicted values. Probabilistic forecasting is interested in learning the full distribution of data, not just a single point realization, to describe the uncertainty in the downstream decision-making (Salinas, Flunkert, Gasthaus, & Januschowski, 2020).

To optimize the probabilistic forecasting, the Gaussian distribution's negative log-likelihood (NLL) is minimized using the loss function $\mathcal{L}_2$. Minimizing the NLL ensures that the predicted distribution is as close as possible to the actual distribution of the target series. This probabilistic approach allows uncertainty quantification in the predictions by providing a full distribution of crude oil prices.

$$\begin{aligned} \mathcal{L}_2 &= \min \sum_{t=1}^{T-1} \frac{1}{2} \log\left(2\pi(\hat{\sigma}^j_{t+1})^2\right) + \frac{1}{2}\left(\frac{\left(x^j_{t+1} - \mu^j_{t+1}\right)^2}{s(\tilde{\sigma}^j_{t+1})^2}\right) + \mathcal{L}_p \\ &= \min \sum_{t=1}^{T-1} \frac{1}{2} \log\left(2\pi(\hat{\sigma}^j_{t+1})^2\right) + \frac{1}{2}\left(\frac{\left(x^j_{t+1} - \hat{\mu}^j_{t+1}\right)^2}{s(\tilde{\sigma}^j_{t+1})^2}\right) + \lambda \sum_{i=1}^{D} \|\mathbf{W}^{:,i}\|_2 \end{aligned} \tag{9}$$

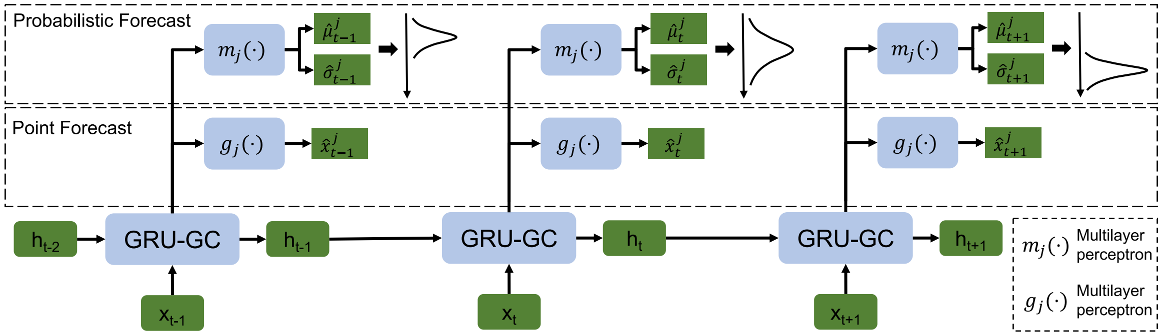


Fig. 3. Two proposed processes of forecasting crude oil price by GRU-GC.

where the values of $\hat{\mu}^j_{t+1}$ and $\tilde{\sigma}^j_{t+1}$ are determined by a feed-forward neural network, denoted as $m_j(h_t)$, which takes the hidden state h_t at time step t as input. Notably, the Softplus activation function is employed to ensure that $\hat{\sigma}^j_{t+1}$, obtained by transforming $\tilde{\sigma}^j_{t+1}$, is strictly positive. Specifically, $\hat{\sigma}^j_{t+1} = s(\tilde{\sigma}^j_{t+1})$, where $s(\tilde{\sigma}^j_{t+1}) = \log(1 + \exp(\tilde{\sigma}^j_{t+1}))$.

By incorporating the probabilistic approach, the GRU-GC model provides point forecasts and confidence intervals that capture the uncertainty inherent in crude oil market data. This is particularly valuable in decision-making processes that require a comprehensive understanding of the range of possible outcomes.

4. Empirical analysis

4.1. Data description

4.1.1. Dependent variables

Brent crude oil prices are widely recognized as a crucial benchmark in the oil market, accounting for approximately 70% of spot crude oil trading prices. This highlights its significant role in shaping the global oil market. On the other hand, WTI crude oil prices also hold substantial influence, considering the United States' prominent position in the global economy and its status as the world's leading consumer of crude oil (Wei, Zhang, & Wang, 2022). Therefore, we have selected Brent and WTI oil spot prices as dependent variables to capture international crude oil price dynamics.

The Brent and WTI oil spot prices are collected from the Wind database, and the statistical descriptions are provided in Tables 2 and 3. In addition to the raw price levels, we also calculate the returns of the Brent crude oil spot price. The formula for computing the return R_t based on the price P_t at time t and the price P_{t-1} at the previous time period is:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{\Delta P_t}{P_{t-1}}.$$

The statistical descriptions of Brent and WTI crude oil prices capture price and return data across training, validation, and test sets, reflecting distinct patterns across these periods. For Brent, the training period shows the highest average price with substantial variability, indicative of significant market fluctuations likely linked to volatile conditions. In contrast, the validation period exhibits the lowest average price with reduced variability, suggesting a phase of relative stability. Skewness and kurtosis in the training set reveal a bias toward downward price movements and a flatter distribution. This pattern shifts in the validation and test sets, where skewness values near zero suggest more balanced price dynamics, and the test set's kurtosis indicates a nearly normal distribution.

In the WTI price dynamics, the training period again shows the highest average price with notable volatility, highlighting significant market activity during this period. The validation period's lower average price and reduced variation reflect greater market stability. For both Brent and WTI, the returns across datasets maintain near-zero mean values, a common characteristic in financial series. Notably, the test set for WTI returns demonstrates exceptionally high kurtosis and skewness, indicating the presence of substantial outliers or price shocks likely tied to unforeseen economic or geopolitical events. This statistical analysis across datasets underscores the necessity of robust forecasting models to effectively capture crude oil's price and volatility dynamics.

Finally, we include a chart to illustrate both spot and futures oil prices to provide a clearer visual understanding of their historical trends and dynamics. Fig. 4 highlights the significant variability in the spot and future price of Brent/WTI crude oil, characterized by substantial fluctuations in both its level and direction. The range of these fluctuations was substantial, with the maximum price reaching 133.18 on March 8, 2022, and the minimum price falling to 9.12 on April 21, 2020.

Table 2
The statistical description of brent spot oil price.

	Mean	Std	Skewness	Kurtosis	Minimum	Median	Maximum
Price							
Training set	86.746	27.794	−0.508	−1.219	26.010	99.205	128.140
Validation set	62.449	10.420	0.241	−1.105	43.980	61.460	86.070
Test set	69.640	23.179	0.260	0.002	9.120	67.335	133.180
Total	79.095	26.819	−0.011	−1.230	9.120	76.275	133.180
Return							
Training set	0.000	0.016	0.493	5.895	−0.079	0.000	0.104
Validation set	0.000	0.017	−0.103	1.370	−0.062	0.000	0.069
Test set	0.001	0.039	0.690	55.542	−0.475	0.002	0.510
Total	0.000	0.025	0.821	97.424	−0.475	0.000	0.510

Table 3
The statistical description of WTI spot oil price.

	Mean	Std	Skewness	Kurtosis	Minimum	Median	Maximum
Price							
Training set	78.919	22.917	−0.615	−1.030	26.190	86.580	113.390
Validation Set	57.671	8.930	0.284	−1.243	42.480	55.210	77.410
Test set	65.026	22.146	0.257	0.150	−36.980	61.375	123.640
Total	72.506	22.937	−0.109	−1.070	−36.980	73.930	123.640
Return							
Training set	0.000	0.018	0.442	6.039	−0.105	0.000	0.120
Validation Set	0.000	0.017	−0.482	2.118	−0.074	0.001	0.076
Test set	−0.002	0.107	−21.822	595.765	−3.020	0.001	0.531
Total	−0.001	0.057	−38.041	1963.304	−3.020	0.000	0.531

4.1.2. Explanatory variables

In examining the factors influencing crude oil prices, researchers have extensively studied various aspects, including supply and demand dynamics, global economic growth, financial factors, energy markets,

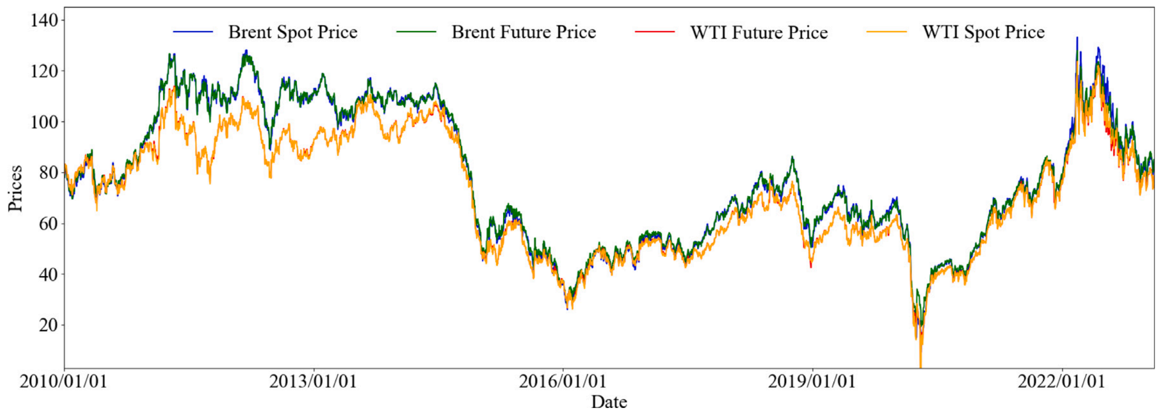


Fig. 4. The visualization of spot and futures with respect to Brent and WTI oil prices.

and external shocks. These factors play a crucial role in understanding the fluctuations in crude oil prices, which are summarized in Table 4.[3]

- **Supply and demand**: The dynamics of supply and demand play a fundamental role in the variability of oil prices. Previous research has demonstrated the significant influence of supply and consumption on oil price fluctuations (Bernabe et al., 2004).
- **Global economic development**: Global economic conditions, reflecting the vitality and well-being of economies, have a profound impact on oil prices, particularly on the demand side.
- **Financial factors**: Financial factors encompass psychological expectations, speculation,[4] and market sentiment that influences oil prices. These factors are measured using data from stock markets and monetary markets.
- **Energy market**: The energy market includes commodities such as natural gas, coal, and steam coal, which are traded on spot and futures markets (Geng et al., 2017; Jin et al., 2023). The prices of these energy commodities are interconnected, with variations in one commodity influencing the demand and price of others.
- **External shocks and uncertainty factors**: The oil market is highly sensitive to external shocks[5] and uncertainty factors[6] that can significantly impact oil prices and variability (Baumeister & Hamilton, 2019; Li et al., 2022; Wei et al., 2017).

In addition, to capture the serial dependencies inherent in the dependent variable, we incorporate historical values of the crude oil spot price into the explanatory variables, specifically using Brent and WTI spot prices from day $t-k$ to t to predict the spot price at day $t+1$. This approach enables the model to effectively account for autocorrelation within the time series, enhancing its capacity to forecast future values. Furthermore, in the context of Granger causality, these historical values of the dependent variable are fundamental to the information set, as their inclusion allows the model to evaluate whether additional covariates contribute incremental predictive power beyond that of the autoregressive components.

[3] Unless otherwise noted, the majority of data utilized in this study originates from the Wind database (ttps://www.wind.com.cn), a widely recognized provider of financial and economic data.

[4] There is no unique definition of speculation in the oil market (Büyükşahin & Harris, 2011; Fattouh, Kilian, & Mahadeva, 2013). The detailed discussion can refer to Kliber, Łęt, and Řezáč (2024).

[5] External shocks include four primary types of shocks from Baumeister and Hamilton (2019).

[6] Uncertainty factor indexes is calculated by Kang and Ratti (2013).

4.2. Benchmark models

To verify the efficacy of the proposed GRU-GC model for forecasting Brent crude oil spot prices, we conduct experiments comparing it with several baseline models as follows.

- ARIMA. It is a generalization of the traditional autoregressive moving average (ARMA) model. It is widely used to comprehend time series data where they are non-stationary in the sense of mean (Hamilton, 2020).
- Naive forecast. It is a fundamental method used in financial asset price forecasting, assuming that today's price is the best estimate for tomorrow's price. In practice, it predicts that the future value of a variable will be equal to its most recent observed value.
- ElasticNet. It is a regularized regression method combining L_1 and L_2 penalties linearly (Hao, Zhao, & Wang, 2020). ElasticNet differs from the proposed GRU-GC, which provides both inter and intra-group regularizations.
- Random Forest Regressor (RF). It is an ensemble learning method that combines several weak learners, i.e., the decision trees, for regression problems (Demirer, Gkillas, Gupta, & Pierdzioch, 2022). Since ensembling some trees can lower the variance of the models, it is used to help forecast time series data.
- Extreme Gradient Boosting for regression (XGBoost). It is a gradient boosting machine implementation (Guliyev & Mustafayev, 2022; Sadorsky, 2022), which can provide automatic feature selection.
- LSTM. It is a standard recurrent neural network that can consider both long-term and short-term memories, making it efficient for handling sequential data (Lu, Sun, Duan, & Wang, 2021).
- GRU. It is the basic unit in the proposed GRU-GC that does not consider any regularization term (Zhang et al., 2023). By comparing them to the basic GRU, we can determine the efficacy of using the inter- and intra-group regularization techniques in time series forecasting problems.
- Temporal Convolutional Network (TCN). TCN can account for the convolutional architectures in the sequential data, which has been proven to achieve state-of-the-art performance across many tasks (Niu, Wang, Lu, Yang and Du, 2021).

4.3. Experimental settings and evaluation metrics

To evaluate the proposed model, we employ the time-based nested cross-validation approach (Tashman, 2000). The data set is split into three parts: a training set covering the period from January 1, 2010, to January 20, 2017; a validation set covering the period from January 21, 2017, to January 20, 2019; and a test set covering the period from January 21, 2019, to January 20, 2023. The training set accounts for

Table 4
Variables related to the crude oil market.

Dependent variables (day $t+1$)	
Oil spot price	Brent oil spot price (USD per barrel), WTI oil spot price (USD per barrel)
Explanatory variables (day $t-k,\ldots,t$)	
Class (Subclass)	Variables (Units)
Supply and demand (Oil production)	U.S. Crude Oil Production (Barrel/day); OPEC Crude Oil Production (Barrel/day); World Crude Oil Production (Barrel/day)
Supply and demand (Oil stock)	World Total Petroleum Stocks (Barrels); U.S. Total Petroleum Stocks (Barrels); U.S. SPR Crude Oil Stocks (Barrels); U.S. Non-SPR Crude Oil Stocks (Barrels)
Supply and demand (Oil consumption)	COMEX Gold Futures Price (USD/oz); LME Copper Futures Prices (USD/tonne); EU HICP on Energy (Percent); U.S. CPI on Energy (Percent); OECD Petroleum Products Consumption (KB/day)
Global economic development	USA PPI Final Demand (Percentile); U.S. PPI on Energy (Percentile); USA Manufacturing PMI (-); U.S. Federal Funds Rate (-); USA 10-year Treasury Yields (-); United States CPI seasonally adjusted (Percentile); EU Harmonized Index of Consumer Prices (Percentile); EU Producer Price Index (Percentile); Normalized Total OECD CLI (-)
Financial factors (Stock market)	The NASDAQ Composite Index (-); S&P 500 Index (-); Dow Jones Industrial Average Index (-); STOXX Europe 50 Index (-);STOXX Europe 600 Index (-); DAX Index of Frankfurt (-); London FTSE 100 Index (-); CAC40 Index of Paris (-)
Financial factors (Monetary market)	Real U.S. Dollar Index (-); EUR to USD Exchange Rate(EUR per USD); USA M1 Money Supply(USD); U.S. M2 Money Supply(USD); Eurozone M1 Money Supply(EUR); Eurozone M2 Money Supply (EUR); Eurozone Benchmark Rate (Percentile); Eurozone 10 Year Bonds (Percentile)
Energy market	Henry Hub Natural Gas Spot Price (USD per million Btu); IPE Rotterdam Coal Futures Price (USD per metric ton); European ARA Ports Steam Coal Spot Price (USD per metric ton); OPEC Crude Oil Basket Price (USD per barrel); WTI Crude Oil Futures Price (USD per barrel); WTI Crude Oil Spot Price (USD per barrel); Brent Oil Futures Price (USD per barrel); Brent Crude Oil Spot Price (USD per barrel)
External shocks and uncertainty factors	Oil Supply Shocks (-); Economic Activity Shocks (-); Oil Consumption Demand Shocks (-); Oil Inventory Demand Shocks (-); Global Economic Policy Uncertainty Index (-); U.S. Monetary Policy Uncertainty Index (-); U.S. Economic Policy Uncertainty (-); U.S. Equity Volatility Index (-); U.S Climate Policy Uncertainty (-); Energy-Related Uncertainty Index (-)

70% of the data, the validation set accounts for 10%, and the test set accounts for 20%.

A sliding window approach is employed to generate data samples, with a window size of twenty days. The data of all variables within the window are used to forecast the next day's Brent oil spot price, capturing the price-changing characteristic. The window was then slid by one day, and the process was repeated to generate subsequent data samples. Fig. 5 visually represents the sample generation process, and Fig. 6 presents the procedures of the proposed GRU-GC.

In the proposed GRU-GC model, several hyperparameters need to be determined, including the learning rate, the degrees of L_1 and L_2 penalties, and the number of hidden units. These hyperparameters are tuned through a grid-searching process.[7]

Table 5
The searching space of training hyperparameters.

Hyperparameters	Searching space
Learning rate	{0.0005, 0.001, 0.003, 0.005}
L_1 penalty	{0.03, 0.04, 0.05, 0.06, 0.07, 0.08, 0.12, 0.15, 0.18, 0.2}
L_2 penalty	{0, 0.001, 0.005, 0.01}
Hidden unit size	{16, 32, 64, 128}

Based on the grid search, the optimal hyperparameters are found to be as follows: the L_1 penalty was set to 0.05, the L_2 penalty was set to 0.005, and the size of the hidden unit in the one-layer GRU was set to 32. The learning rate was fixed at 0.001. Table 5 displays the grid of possible values for each hyperparameter. The optimal hyperparameters determined through grid search are as follows: an L_1 penalty of 0.05, an L_2 penalty of 0.005, a hidden unit size of 32 in the one-layer GRU, and a fixed learning rate of 0.001.

During the training process, the results of each epoch are stored, and the model with the best performance is selected for forecasting the test data. To comprehensively evaluate the results, four evaluation metrics are employed for point prediction: root mean square error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE), and R^2. These evaluation metrics are defined as follows:

$$\mathrm{RMSE} = \sqrt{\frac{1}{T}\sum_{t=1}^{T}\left(\hat{x}_t^j - x_t^j\right)^2} \tag{10}$$

$$\mathrm{MAE} = \frac{1}{T}\sum_{t=1}^{T}\left|\hat{x}_t^j - x_t^j\right| \tag{11}$$

$$\mathrm{MAPE} = \frac{1}{T}\sum_{t=1}^{T}\left|\frac{\hat{x}_t^j - x_t^j}{x_t^j}\right| \times 100\% \tag{12}$$

$$\mathrm{R}^2 = 1 - \frac{\sum_{t=1}^{T}\left(x_t^j - \hat{x}_t^j\right)^2}{\sum_{t=1}^{T}\left(x_t^j - \bar{x}^j\right)^2} \tag{13}$$

where $\hat{x}_t^j$ and x_t^j are the predicted and true values of variable X_t^j at the tth time step, $\bar{x}^j$ is the average of the true values over T time steps. RMSE measures the average magnitude of the errors, with larger errors given more weight due to the squaring operation. MAE measures the average magnitude of the errors without considering their directions. MAPE provides a percentage measure of the average absolute percentage error, which is not affected by the scale of the predictions. R^2 measures the proportion of the variance in the true values that are captured by the predicted values. In general, a model with lower RMSE, MAE, MAPE, and higher R^2 is considered to have better predictive performance.

5. Experimental results

5.1. Forecast results

The performance of the proposed GRU-GC model is compared with eight baseline methods. Since the baseline methods do not provide probabilistic forecasts, only the point forecast results for the GRU-GC are presented in Table 6.

For the daily forecasting scenario ($n = 1$), the input consists of the first twenty-day values of the time series, and the target is the next day's Brent oil spot price. The results show that deep neural network-based methods, including TCN, LSTM, and GRU, outperform linear models such as ARIMA and ElasticNet. The enhanced performance can

[7] Grid search operates by setting up a grid of possible values for each hyperparameter. It then trains and evaluates the proposed GRU-GC model using each combination, selecting the configuration that yields the best performance. We measure point and probabilistic forecasting accuracy using root mean square error (RMSE) and NLL, respectively.

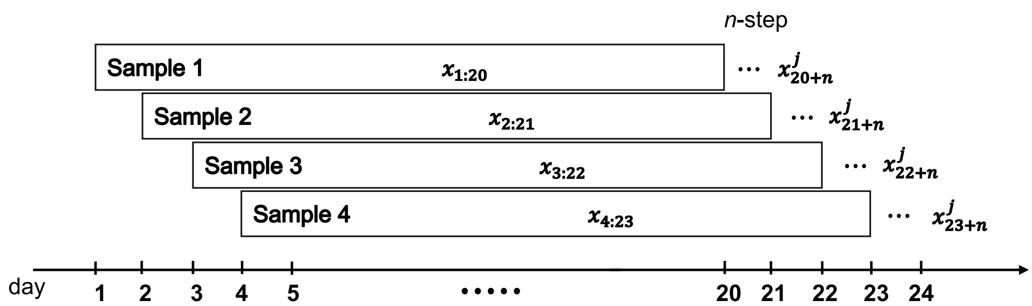


Fig. 5. Time-based sample generation process. Note that there is a n-step lag between the training and forecasting data points and $n \in \{1, 5, 10\}$.

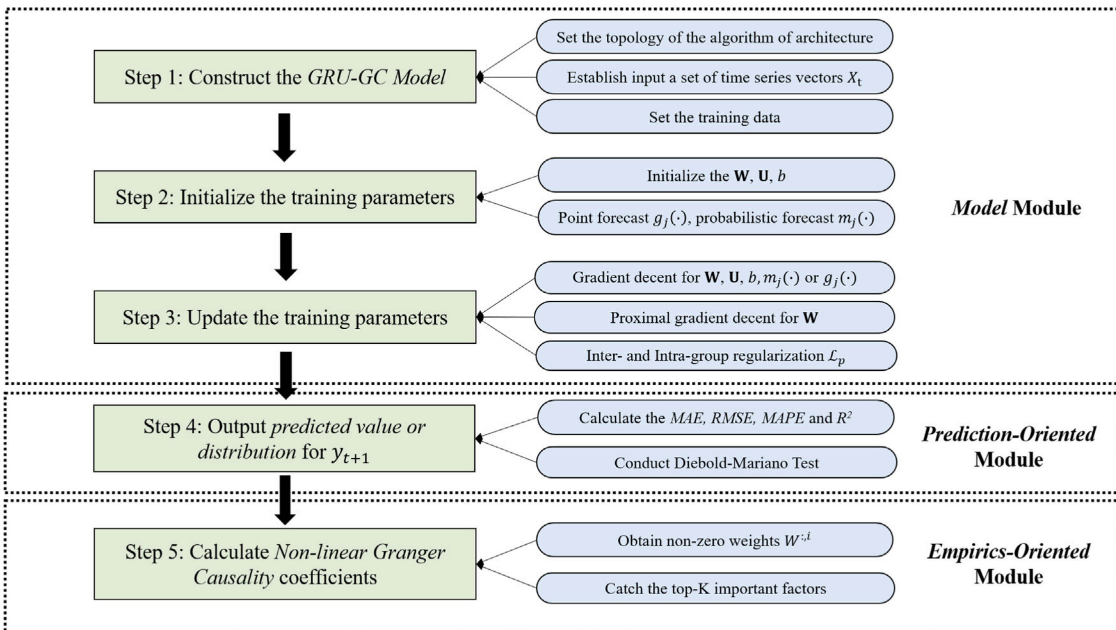


Fig. 6. The procedures of the proposed GRU-GC.

Table 6
The experimental point forecast results on n-step ($n = 1, 5, 10$) ahead forecast.

Models	Step $n = 1$				Step $n = 5$				Step $n = 10$			
	RMSE	MAE	MAPE	R^2	RMSE	MAE	MAPE	R^2	RMSE	MAE	MAPE	R^2
ARIMA	3.025	2.139	3.402%	0.983	4.930	3.481	5.695%	0.956	6.175	4.507	7.758%	0.932
Naive forecast	2.401	1.495	2.631%	0.990	4.479	3.067	5.042%	0.963	5.887	4.209	7.194%	0.936
ElasticNet	13.948	8.582	10.675%	0.642	15.548	9.744	12.904%	0.557	20.559	14.192	19.784%	0.228
XGBoost	2.944	1.897	4.162%	0.984	6.256	4.801	8.993%	0.928	8.191	6.518	12.232%	0.878
RF	3.021	1.945	4.285%	0.983	6.230	4.518	8.909%	0.929	8.559	6.558	12.474%	0.866
LSTM	4.912	2.621	3.718%	0.955	5.826	3.972	6.288%	0.937	8.801	6.683	10.952%	0.856
GRU	2.799	1.833	2.891%	0.985	5.066	3.305	5.050%	0.952	9.055	6.986	11.838%	0.847
TCN	2.448	1.507	2.648%	0.989	4.653	3.224	5.100%	0.958	6.074	4.261	**6.644%**	0.927
GRU-GC	**2.122**	**1.391**	**2.563%**	**0.992**	**3.164**	**2.452**	**4.042%**	**0.986**	**5.033**	**4.172**	6.510%	**0.956**

be attributed to the capacity of deep neural networks to model complex, nonlinear relationships among input variables effectively. The proposed GRU-GC model outperforms the compared methods in terms of MAE, RMSE, MAPE. This superior performance is attributed to the model's ability to penalize irrelevant variables, prevent overfitting, and capture nonlinear relationships, making it highly effective in forecasting Brent oil spot prices.

Apart from model-based forecasting, we find that the performance of the naive forecast approach outperforms most model-based models. It is reasonable since financial markets are often characterized by the random walk hypothesis, which posits that asset prices evolve according to a random path and all available information is already reflected in current prices (Miller, Roll, Taylor, et al., 1970). Furthermore, Complex forecasting methods often involve numerous parameters and assumptions, increasing the risk of overfitting historical data and capturing noise rather than meaningful patterns (Chatfield, 1995). Such models can become fragile, performing poorly when market conditions change or when applied to new data sets. Naive forecasting avoids these pitfalls by making minimal assumptions about the data and relying solely on the most recent observation for prediction.

In addition to the one-step ahead forecast, we extend the forecasting step to $n = 5$ and $n = 10$ to evaluate the proposed GRU-GC model's performance in forecasting weekly and bi-weekly Brent oil spot prices. This longer-term result helps to assess the model's robustness during significant events or policy changes. As the lag increases, the predictive accuracy of both the GRU-GC and baseline methods declines, as indicated by higher MAE, RMSE, MAPE. Despite this decrease, the GRU-GC

Table 7
The results of pairwise DM tests.

	Step $n = 1$	Step $n = 5$	Step $n = 10$
ARIMA	-7.165^{***}	-6.621^{***}	-3.568^{***}
Naive forecast	-4.002^{***}	-6.294^{***}	-2.855^{**}
ElasticNet	-17.130^{***}	-16.774^{***}	-17.287^{***}
RF	-6.316^{***}	-11.715^{***}	-12.395^{***}
LSTM	-14.206^{***}	-12.783^{***}	-12.020^{***}
XGBoost	-6.251^{***}	-13.632^{***}	-11.760^{***}
TCN	-4.375^{***}	-6.975^{***}	-3.002^{**}
GRU	-4.881^{***}	-7.070^{***}	-16.090^{***}

Note: Significant level *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

model consistently outperforms the baselines in these metrics. Furthermore, while the R^2 value decreases with longer lags-dropping by 1% for $n = 5$ and 4.14% for $n = 10$-the decline is minimal, demonstrating the GRU-GC's robustness in capturing the overall price-changing pattern better than the baselines.

Generally, model performance is assessed using error metrics, but this method can be unreliable due to poor sampling. To address this, the Diebold–Mariano (DM) test evaluates significant performance differences between models (Diebold & Mariano, 2002). The null hypothesis of the DM test asserts that the GRU-GC model and a baseline model perform equally. The alternative hypothesis suggests a significant performance difference. The DM test results are shown in Table 7, with a larger DM test statistic value signifying a difference. Conducting the DM test provides statistical evidence of the GRU-GC model's superior forecasting accuracy over the baselines.

From Fig. 7(a) and (b), it is evident that the proposed GRU-GC model, both in point and probabilistic forecasts, demonstrates superior performance compared to the ARIMA and GRU models in approximating the oil price trend. The point forecast GRU-GC exhibits closer proximity to the actual oil price curve, indicating its effectiveness in capturing the underlying patterns and dynamics of the oil market.

However, during the period from February 2020 to June 2020, there is a noticeable disparity between the predicted and actual oil prices, with a sharp decline in the latter. The observed discrepancy is largely due to the effects of the COVID-19 pandemic on the global oil market. The pandemic disrupted both supply and demand, resulting in significant fluctuations in oil prices, as illustrated by the study (Bouazizi, Guesmi, Galariotis, & Vigne, 2024). The point forecast methods, including the proposed GRU-GC, may face limitations in capturing such extreme events and outliers.

In contrast, the first advantage of the probabilistic forecasting approach, as shown in Fig. 7(b), is providing a price or confidence interval that encompasses the uncertainty and variability of oil prices. Unlike point forecasts, which provide a single realization, probabilistic forecasts are interested in learning a full distribution and offer a range of possible outcomes, including some unexpected events such as the breakout of pandemics. This helps policymakers understand the level of uncertainty in their predictions and make more informed decisions (Salinas et al., 2020).

Another fundamental advantage of probabilistic forecasting is its ability to incorporate human prior knowledge into decision-making by assuming data distribution. As illustrated in Fig. 7, although we have assumed the oil price follows the Gaussian distribution, it can be replaced by any distribution once the policymakers can describe the patterns based on either his/her experiences or historical data. Therefore, it enables policymakers to manage risks more effectively and ensures the decisions align with the human experiences (Gneiting & Katzfuss, 2014).

Overall, while the point forecast GRU-GC demonstrates superior performance in approximating the oil price trend, the probabilistic forecast GRU-GC is recommended for practical applications, particularly in periods of high variability and uncertainty, as it provides policymakers with a more comprehensive understanding of the oil price distribution and associated confidence intervals.

5.2. Causality results

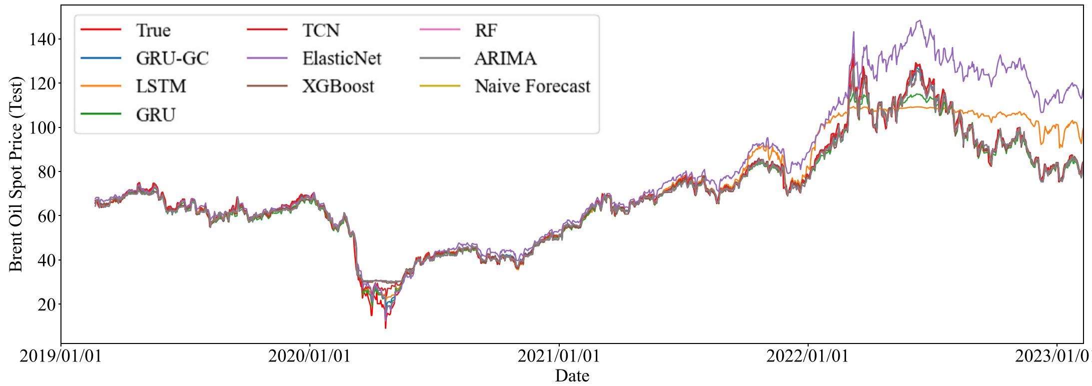
As shown in Eq. (8), the non-zero weights $\mathbf{W}^{:,i}$ indicate Granger causality of ith series to the target series x_t^j. To compare and confirm the causal factors of crude oil prices identified by the GRU-GC model, we supplemented the existing Brent crude oil data with WTI crude oil data. Specifically, there are a total of 36 and 33 non-zero weights of factors for Brent and WTI oil price prediction, respectively. To get deeper insights, we further extract the three most significant factors from the identified factors, as illustrated in Table 8.

Table 8 presents the ranking of various factors based on their importance in predicting Brent and WTI crude oil prices. These factors are grouped into categories, including the energy market, global economic development, financial factors, demand and supply, external shocks, and uncertainty factors. Within each category, factors are listed from most to least influential on oil prices. The results indicate that both Brent and WTI prices are predominantly influenced by Energy Market factors, emphasizing the direct role of indicators like spot and futures prices in short-term forecasting. Global Economic Development and Financial Factors also play a significant role, suggesting that macroeconomic conditions and financial metrics — such as stock and monetary indicators — strongly affect crude oil prices. Categories like Demand and Supply, along with External Shocks, show moderate importance, highlighting how production levels, consumption, and unforeseen disruptions impact prices. Lastly, Uncertainty Factors, particularly those related to policy and economic volatility, hold influence as well, albeit ranking lower, reflecting the indirect yet significant effects of geopolitical and economic uncertainties on market expectations. This categorization offers a comprehensive perspective on oil price fluctuations' primary economic and market drivers. In the following sections, we thoroughly examine the specific indices within each class.

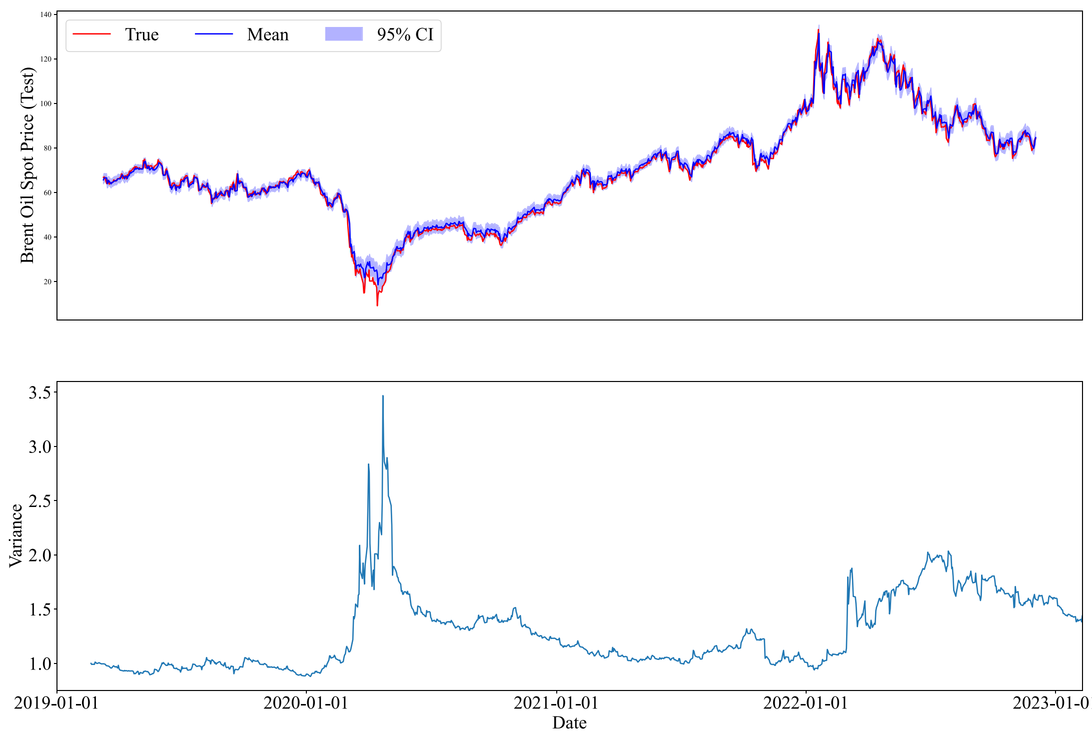
5.2.1. Energy market

In the energy market, the findings in Table 8 emphasize the significant roles of spot and futures prices for both Brent and WTI crude oil, underscoring the importance of these variables in price forecasting. For the Brent market, the Brent Crude Oil Spot Price exerts the most substantial influence, followed by the Brent Oil Futures Price and then the WTI Crude Oil Futures Price. This ordering suggests that while Brent-specific indicators hold primary relevance, global factors represented by WTI futures still bear considerable weight. In contrast, the WTI market shows the highest sensitivity to the WTI crude oil spot and futures prices, with the European ARA ports' steam coal spot price also emerging as a significant factor. The inclusion of this coal price for WTI highlights the interconnected structure of the energy commodities market, where pricing dynamics in one sector (e.g., coal) impact others, reflecting substitution effects or broader shifts in energy demand. This interrelation across energy commodities suggests that coal, natural gas, and oil markets are jointly influenced by overlapping supply–demand fundamentals and policy factors, particularly in energy-intensive economies.

These dynamics reveal a complex interdependence between spot and futures markets, with futures prices in both Brent and WTI markets acting as forward-looking indicators that significantly impact spot price movements. This relationship is well-documented in financial literature, which notes that futures prices can influence spot prices by incorporating market expectations, speculative behaviors, and anticipated supply–demand changes (Huang, Yang, & Hwang, 2009; Mamatzakis & Remoundos, 2010). However, capturing this relationship poses a challenge due to its nonlinearity and time-varying nature, which traditional econometric models often fail to represent fully. By employing the GRU-GC model, our analysis is better equipped to capture these complex predictive dynamics.



(a) The point forecast results of GRU-GC on the test data set.



(b) The probabilistic forecast results of GRU-GC on the test data set.

Fig. 7. The forecast results provided by point and probabilistic forecasts. (best viewed in color). In top plot of panel (b), the shaded region represents the 95% confidence interval, computed as $CI = \hat{\mu}_t \pm 1.96\hat{\sigma}_t$, where $\hat{\mu}_t$ and $\hat{\sigma}_t$ are derived from the process illustrated in Fig. 3. The bottom plot in panel (b) displays the predicted variance $\hat{\sigma}_t^2$, which quantifies the model's estimated uncertainty at each time step.

5.2.2. *Supply and demand*

Our analysis reaffirms the fundamental role of both supply and demand in driving crude oil price fluctuation. Demand-side indicators, such as the U.S. CPI on Energy, LME Copper Futures Prices, and the EU HICP on Energy, prove especially critical for Brent crude, as they mirror broader economic activity and shifts in energy demand. For WTI, similar demand factors emerge, with the LME Copper Futures Prices, COMEX Gold Futures Prices, and EU HICP on Energy identified as influential metrics. These demand indicators act as proxies for overall economic health, reflecting trends in consumption and energy requirements, which, in turn, directly impact oil price variability.

On the supply side, our findings reveal that key production metrics, including OPEC Crude Oil Production, World Crude Oil Production, and U.S. Crude Oil Production, significantly affect Brent prices by shifting the global supply balance. For WTI, production influences are similarly rooted in U.S., OPEC, and global production levels, reflecting the interconnected structure of international oil markets. The interaction between these production metrics and global demand creates a dynamic supply environment that constantly adjusts to maintain market equilibrium.

Inventory levels add a further layer of complexity to the oil price structure by functioning as a buffer that moderates the effects of production and demand fluctuations on market stability. Changes in inventory levels can significantly alter supply–demand dynamics, influencing short-term price movements as stocks absorb production surpluses or counterbalance shortages. The strategic role of inventories in stabilizing the market, highlighted in studies by Hamilton (1983) and Lu et al. (2020), is crucial for understanding how markets adjust to supply shocks and demand surges. By accommodating fluctuations in production or consumption, inventories stabilize prices, mitigating extreme variability and contributing to long-term market resilience. Supply, demand, and inventory factors collectively shape the crude oil

Table 8
The experimental results of one-step ahead Granger causality forecast. (Note that > indicates the relative importance of two factors).

Panel A: Ranking of factors for brent	
Class (from important to insignificant)	Top 3 Factors
Energy market	Brent Crude Oil Spot Price > Brent Oil Futures Price > WTI Crude Oil Futures Price
Global economic development	OECD Petroleum Products Consumption > USA: Federal Funds Rate > USA PPI on Energy
Financial factors(Stock market)	STOXX Europe 50 Index > CAC40 Index of Paris > DAX Index of Frankfurt
Oil stocks	U.S. Non-SPR Crude Oil Stocks > U.S. SPR Crude Oil Stocks > World Total Petroleum Stocks
Demand	U.S. CPI on Energy > LME Copper Futures Prices > EU HICP on Energy
Financial factors(Monetary market)	Eurozone Benchmark Rate > Eurozone 10 Year Bonds > Real U.S. Dollar Index
Supply	OPEC Crude Oil Production > World Crude Oil Production > U.S. Crude Oil Production
External shocks	Oil Supply Shocks > Economic Activity shocks > Oil Consumption Demand Shocks
Uncertainty factors	U.S Climate Policy Uncertainty > Energy-Related Uncertainty Index > Global Economic Policy Uncertainty Index
Panel B: Ranking of factors for WTI	
Class (from important to insignificant)	Top 3 Factors
Energy market	WTI Crude Oil Spot Price > WTI Crude Oil Futures Price > European ARA Ports Steam Coal Spot Price
Global economic development	OECD Petroleum Products Consumption > USA: Federal Funds Rate > USA: 10-year Treasury Yields
Financial factors(Monetary market)	U.S. M2 Money Supply > USA M1 Money Supply > Eurozone Benchmark Rate
Demand	LME Copper Futures Prices > COMEX Gold Futures Price > EU HICP on Energy
Financial factors(Stock market)	STOXX Europe 600 Index > STOXX Europe 50 Index > London FTSE 100 Index
Oil stocks	U.S. Non-SPR Crude Oil Stocks > U.S. Total Petroleum Stocks > U.S. SPR Crude Oil Stocks
Supply	U.S. Crude Oil Production > OPEC Crude Oil Production > World Crude Oil Production
External shocks	Oil Supply Shocks > Economic Activity shocks > Oil Consumption Demand Shocks
Uncertainty factors	U.S Climate Policy Uncertainty > Energy-Related Uncertainty Index > Global Economic Policy Uncertainty Index

pricing landscape, underscoring the need for models that can account for these intertwined elements to accurately predict price variability.

5.2.3. Global economic development

Our analysis demonstrates that global economic development also significantly influences crude oil prices, with the OECD Petroleum Products Consumption and the U.S. Federal Funds Rate emerging as critical determinants for both Brent and WTI markets. In the case of Brent, these indicators, along with the U.S. PPI on Energy, establish a framework in which global economic health and growth cycles directly shape price fluctuations in the oil market. For WTI, the combination of the U.S. 10-year Treasury Yields with the Federal Funds Rate and OECD consumption metrics suggests an acute sensitivity to U.S. monetary policy and long-term fiscal signals, underscoring the role of U.S. economic expectations in influencing oil demand and prices.

The critical role of these economic indicators aligns with findings by Lu et al. (2020), which emphasize the strong relationship between global economic activity and oil price dynamics. OECD Petroleum Products Consumption serves as a barometer of global economic conditions, reflecting consumption trends in leading economies and thus directly influencing oil demand and price movements. On the other hand, the U.S. Federal Funds Rate has far-reaching implications for investment climates and the relative strength of the U.S. dollar. Since oil is priced in dollars, shifts in the dollar's value alter the purchasing power of oil-importing countries, indirectly affecting global demand. Meanwhile, the U.S. PPI on Energy and the 10-year Treasury Yields offer insights into inflationary pressures and long-term economic forecasts, which are vital for understanding anticipated future costs and returns in the oil markets. The inclusion of these indicators in our model supports the argument that crude oil markets are not only responsive to current economic activity but are also intricately connected to fiscal policies, inflation expectations, and other forward-looking economic indicators. This complex interplay between economic health, fiscal signals, and inflation trends underscores the broader economic forces that drive volatility and pricing in the global oil market, making these indicators essential for a comprehensive analysis of oil price fluctuations.

5.2.4. Stock market

Our analysis underscores the influence of stock market indices on crude oil prices, highlighting the interconnectedness between global equity markets and commodities. For Brent crude, the STOXX Europe 50, CAC40 Index of Paris, and DAX Index of Frankfurt emerge as key indices, serving as indicators of economic stability and investor confidence across major European economies. These indices reflect broader economic conditions and sentiment within Europe, directly impacting demand for energy commodities like crude oil. In the WTI market, the STOXX Europe 600, STOXX Europe 50, and London FTSE 100 hold substantial influence, indicating that broader European and UK financial trends also play a critical role in shaping expectations and demand patterns for WTI. The inclusion of these European stock indices highlights the far-reaching effects of regional economic health on global oil demand, particularly as European markets serve as economic bellwethers that signal broader demand trends within the energy sector.

These findings align with the literature on the dynamic relationship between stock markets and oil prices, where equity indices act as proxies for economic growth, financial stability, and investor sentiment. Strong performance in equity markets often indicates a thriving economy, driving up demand for oil as industrial production and consumption rise. Conversely, downturns in these indices may signal economic contraction, prompting reduced oil demand. Additionally, stock markets reflect investor sentiment and risk appetite, influencing speculative behavior in oil markets as investors adjust their portfolios based on economic outlooks. The presence of these indices in our analysis not only demonstrates the regional dependencies of Brent and WTI but also illustrates how shifts in financial markets can reverberate through commodity markets. Understanding this interdependence is crucial for policymakers and investors alike, as it enables them to better anticipate oil price dynamics based on broader financial and economic trends.

5.2.5. Monetary market

Our analysis reveals that monetary factors, including interest rates, money supply, and exchange rates, play a significant role in shaping crude oil price dynamics in both the Brent and WTI markets. For Brent crude, influential variables include Europe's benchmark interest rate, 10-year government bond yields, and the Real U.S. Dollar Index. These factors are pivotal in determining currency valuations, which in turn impact oil prices by altering the relative purchasing power of international buyers. For WTI, the analysis highlights the U.S. money supply and Europe's benchmark rate as key determinants. The U.S. money supply, managed by the Federal Reserve, directly influences the dollar's value, which is critical in a globally traded commodity like oil, as fluctuations in the dollar's strength alter the cost of oil in other currencies. This shift impacts international demand for WTI, driving either increases or decreases in its price. Europe's benchmark rate also plays a dual role, affecting the euro's value, and thereby impacting both Brent and WTI similarly, as changes in the euro-dollar exchange rate indirectly shape global oil price dynamics.

The interaction of these monetary factors unveils a complex framework in which interest rates and money supply influence financial markets and commodity markets like crude oil. Crude oil prices are highly sensitive to shifts in monetary policy, reflecting how central bank decisions in major economies affect investment environments, economic activity, and global oil demand. Higher interest rates, for instance, tend to strengthen the currency, reduce inflationary pressures, and potentially dampen oil demand by slowing economic activity. Conversely, an expansive money supply can weaken the dollar, making oil more affordable internationally and thereby boosting demand. Through our GRU-GC model, we are able to capture these intricate relationships and the broader economic forces driving oil prices, offering a more comprehensive view of how monetary policy decisions reverberate through the global economy to impact oil markets.

5.2.6. External shocks and uncertainty factors

External shocks and uncertainty factors emerge as highly influential drivers of crude oil price variability, reflecting their potential to disrupt market stability and create significant price shifts in response to unpredictable events, oil supply shocks — often arising from geopolitical tensions, conflicts, or sudden adjustments in production quotas by major oil-exporting countries — have the most pronounced effect (Baumeister & Hamilton, 2019; Li et al., 2022; Zhang, Ji, Zhang and Guo, 2024). These shocks directly alter the supply chain, causing immediate adjustments in prices as markets respond to potential or actual disruptions in oil availability. Economic activity shocks, such as global financial crises or rapid economic expansions, also impact Brent prices by shifting global demand and influencing investment levels in the energy sector. Additionally, oil consumption demand shocks, whether driven by changes in consumer behavior or technological advancements in energy efficiency, further impact Brent prices by altering demand dynamics. For WTI, supply shocks similarly hold the most significant influence, followed by economic activity and consumption demand shocks, which together shape WTI's price variability in response to sudden economic or supply chain disruptions.

The interplay of these external shocks highlights a complex system in which unexpected changes — be they geopolitical, economic, or technological — can destabilize the oil market, triggering substantial price movements. This hierarchical effect of shocks aligns with findings from Baumeister and Hamilton (2019) and supports the robustness of the GRU-GC model in capturing these impacts. The GRU-GC model's ability to detect and interpret these shocks enhances our understanding of crude oil price dynamics, allowing market participants to anticipate potential price movements and adjust their strategies accordingly. By analyzing the mechanisms through which these shocks propagate, the model provides insights into how price volatility can arise in response to various types of market disruptions.

Within the broader category of uncertainty factors, we identify three critical indices that significantly affect both Brent and WTI: the U.S. Climate Policy Uncertainty Index, the Energy-Related Uncertainty Index, and the Global Economic Policy Uncertainty Index. The prominence of the U.S. Climate Policy Uncertainty Index and the Energy-Related Uncertainty Index underscores the influential role of U.S. energy and environmental policies in shaping global oil markets. Given the U.S.'s substantial presence in oil production and consumption, policy shifts related to climate and energy carry substantial weight in influencing market expectations and risk assessments. The Global Economic Policy Uncertainty Index also ranks as a significant factor, capturing the wider impact of global economic policy shifts on oil market sentiment and investment flows.

6. Conclusion

The accurate forecasting of crude oil prices is indeed of great importance for policymakers, financial investors, and researchers due to its significant impact on the global economy. For instance, policymakers can use these insights to anticipate economic impacts and design more responsive policies to manage energy costs and inflationary pressures. Financial investors and analysts benefit by incorporating predictive insights into their investment strategies, helping to mitigate risks associated with oil price volatility. Additionally, this work offers researchers a framework that leverages advanced machine learning techniques to uncover and analyze nonlinear relationships, contributing to the field's understanding of oil market dynamics. The implications of oil price forecasting are wide-ranging and can provide insights into the underlying dynamics of the oil market, informing policy decisions and risk management strategies.

In this study, the proposed GRU-GC model demonstrates its effectiveness in point predictions of Brent crude oil prices. The model outperforms benchmark models, including statistical and machine learning approaches, highlighting the importance of capturing nonlinear relationships among various factors. The integration of deep neural networks and the Granger causality concept through group lasso regularization enables the identification of significant causal relationships in a group manner, leading to improved predictive accuracy.

Furthermore, the GRU-GC model extends its capabilities beyond point forecasting by fitting a Gaussian distribution to the target series. This allows for the estimation of probabilistic predictions, particularly valuable in situations of high uncertainty. The probabilistic predictions provide a range of possible outcomes, providing policymakers with a more comprehensive understanding of the potential price fluctuations.

The limitations of the study are discussed as follows. First, in the case of linear Granger causality, the coefficients associated with covariates offer direct insights into the nature of each covariate's relationship with the target variable. However, the GRU-GC model operates in a fully non-linear domain, where the relationships between covariates and the target variable are more intricate. Although the relative importance of each covariate can be inferred from the magnitude of the weight matrix W, the non-linear transformations involved complicate a straightforward interpretation of coefficients, particularly in terms of direction. Second, the GRU-GC model is optimized for variables that share the same observation frequency, such as daily or monthly intervals. However, in scenarios involving mixed-frequency data, like those often encountered in the energy market, some macroeconomic indicators are recorded weekly or monthly. When adapting GRU-GC to handle such mixed-frequency data by filling in low-frequency variables with repeated values, the information content of these variables is diluted. This reduction in information can diminish their predictive power with respect to the target variable, leading to a potential misrepresentation of Granger causality relationships. Consequently, the model might underestimate or misidentify the influence of these mixed-frequency variables, potentially impacting the accuracy of causality inferences. Finally, the analysis is based on a limited data set and period, which may only partially capture some market conditions and potential variations in the relationships between variables over time. Also, further research is needed to validate the findings and explore the applicability of the GRU-GC model in other domains beyond crude oil price forecasting.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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